

**“SnackSwap: Reverse Vending for Plastic Bottles and Cans with Monitoring for
Supporting the Community”**

by

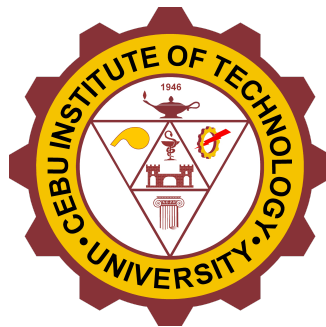
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of the Requirements for the Degree
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ABSTRACT

SnackSwap: Reverse Vending for Plastic Bottles and Cans with Monitoring for Supporting the Community is a prototype system that encourages recycling through rewards. It allows users to deposit 500ml transparent plastic bottles or aluminum cans in exchange for snacks or water, promoting both environmental care and community well-being.

The machine uses a combination of capacitive, inductive, and load sensors to accurately identify valid recyclables. A mobile app, developed with Blynk, remotely monitors trash bin levels and water tank status with a manual button that controls data uploads to stay within the 30,000-point limit of the Blynk Cloud.

Testing showed 100% accuracy in identifying bottles and cans, using a weight range of 2.4–6g to reject invalid items like wood or spray cans. Ultrasonic sensors tracked bin fill levels with 95–99% accuracy and an average response time of 11.77 seconds. Reward dispensing worked flawlessly in repeated tests. A user survey reported high satisfaction, with usability scores between 4.3 and 5.0 out of 5.

Overall, SnackSwap proved to be accurate, reliable, and user-friendly. Future upgrades could include accepting more types of recyclables, improving sensor calibration, expanding reward options, and adding environmental education features to the system.

ACKNOWLEDGMENTS

We want to thank everyone who helped us with this project. From the skies above, we thank God for giving us this opportunity and the perseverance to pursue this project. His guidance continues to give us strength and purpose as we move forward. We are also grateful to our adviser, Engr. Jundith Alterado, for her guidance and support. Her advice has been valuable throughout this process. Our heartfelt thanks go to our families and friends for their encouragement and understanding. Their support gives us the motivation to continue.

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM BACKGROUND

The improper disposal of plastic bottles and aluminum cans is a common issue in many communities. These materials are often left scattered around, creating untidy surroundings and contributing to waste problems. One of the reasons this happens is that people are not fully aware of the importance of recycling and its benefits to the community. According to Becerril-Arreola and Bucklin (2020), beverage container waste is a significant contributor to environmental pollution, accounting for millions of wasted recyclable materials annually. This affects community cleanliness and results in the loss of valuable resources and energy, which could have been conserved through proper recycling practices.

The SnackSwap: Reverse Vending for Plastic Bottles and Cans with Monitoring for Supporting the Community project addresses this issue by raising awareness about recycling while offering tangible incentives. By allowing individuals to exchange recyclable materials like transparent plastic bottles and aluminum cans for snacks or drinking water, this initiative encourages proper waste disposal and contributes to environmental sustainability. The integration of advanced sensor technologies ensures accurate material verification, and monitors inventory levels. This community-focused approach not only promotes recycling but also provides basic necessities to vulnerable populations, making recycling an accessible and rewarding activity.

The project, SnackSwap: Reverse Vending for Plastic Bottles and Cans with Monitoring for Supporting the Community, aims to address this problem by raising awareness about recycling. By allowing people to exchange recyclable materials like plastic bottles and aluminum

cans for snacks or drinking water, this project provides a small reward to encourage proper waste disposal.

A study by Amantayeva et al. (2021) investigated the integration of reverse vending machines (RVMs) into Kazakhstan's waste management system. Through a comprehensive public survey, the study identified key factors influencing recycling behaviors, including public awareness, willingness to engage with RVMs, and the effectiveness of incentives. The findings revealed that appropriate incentives significantly motivated individuals to participate in recycling programs.

Similarly, research by Singh and Kumari (2019) examined the challenges and adoption of RVMs in India. Their study highlighted that offering monetary rewards, such as coupons or e-wallet credits, for recycling plastic bottles through RVMs effectively increased public participation in recycling initiatives.

These studies underscore the potential effectiveness of the SnackSwap project in promoting recycling behaviors by providing tangible incentives, thereby fostering environmental sustainability and community engagement.

The system will also include a feature to track the status of the machine, helping to monitor its operation and results. Through this approach, the project seeks to inform and motivate individuals about the benefits of recycling, helping to create cleaner and more responsible communities.

1.2 PROJECT GOAL AND OBJECTIVES

Project Goal

This project focuses on creating a system that promotes recycling and raises awareness about proper waste disposal. It allows people to exchange items plastic bottles and aluminum cans for snacks or drinking water. The system is designed to encourage recycling practices within the community and reduce waste.

Objectives

To achieve this goal, the project aims:

1. To be able to design and build a reverse vending machine that accepts plastic bottles and aluminum cans.
2. To provide a reward system that exchanges recyclable items for snacks or drinking water.
3. To monitor water level and trash level, on plastic bottle bin and aluminum can bin, through App.

1.3 SIGNIFICANCE OF THE STUDY

1.3.1. Environmental Impact

By encouraging recycling activity, the project aims to help reduce waste and conserve resources, contributing to a cleaner and healthier environment.

1.3.2. Community Involvement

By providing rewards for people to take part in recycling activity, the project promotes a feeling of community involvement.

1.3.3. Social Impact

By giving incentives like water and snacks, the project directly helps low-income and homeless people, among other vulnerable groups, by addressing their basic necessities.

1.4 THEORETICAL BACKGROUND

The SnackSwap: Reverse Vending for Plastic Bottles and Cans with Monitoring for Supporting the Community project is based on four key theories: behavioral reinforcement, community participation, feedback mechanisms, and environmental education. These theories guide the project's design and inform how it can help improve recycling habits within the community.

1.4.1. Behavioral Reinforcement

According to B.F. Skinner's work in "The Behavior of Organisms: An Experimental Analysis" (1938) introduced the concept of operant conditioning, which explains how rewards or punishments influence behavior.

Skinner demonstrated that when a behavior is followed by a positive result, such as a reward, it is more likely that the behavior will be repeated in the future. For example, in the context of the SnackSwap project, people who recycle plastic bottles or aluminum cans receive rewards like snacks or drinking water. The reward acts as reinforcement, encouraging individuals to recycle more frequently.

This concept of reinforcement is central to SnackSwap. When individuals are rewarded for their recycling efforts, they are more likely to see recycling as a beneficial activity and continue doing it regularly. Over time, the act of recycling becomes linked to a positive experience, helping to build a habit. This process turns recycling from an occasional or one-time activity into a consistent behavior. Skinner's theory shows how rewards not only influence behavior but can shape it over time. Through SnackSwap, the project aims to reinforce recycling habits, making it a regular part of daily life for individuals in the community.

In the case of SnackSwap, the rewards of snacks and water serve as immediate and tangible incentives that encourage individuals to engage in recycling. The goal is to build a routine where recycling becomes second nature for participants, and over time, the need for external rewards may lessen as the behavior becomes more ingrained.

1.4.2. Community Participation

Community involvement plays a key role in the success of recycling programs. Research by Gainforth et al. (2016) in “Developing Interventions to Promote Recycling: A Systematic Review of Applicability to Behavior Change Frameworks” shows that people are more likely to participate in recycling when they feel part of the process and can see the benefits. The SnackSwap project uses this idea by offering rewards like snacks and drinking water for recycling, which makes recycling more appealing and encourages people to recycle more often.

Although SnackSwap participants do not see the data on their recycling directly, the project has a monitoring that helps administrators track important details, such as trash levels and the water levels in the vending machines, making sure the machines are always ready for use.

1.4.3. Feedback Mechanisms

Providing feedback is an effective way to encourage and maintain recycling behavior. Schultz’s study in “Changing Behavior With Normative Feedback Interventions: A Field Experiment on Curbside Recycling” (1999) found that people are more likely to recycle when they receive feedback about their recycling efforts, especially when they can compare their behavior to others in their community. When individuals are made aware of how their recycling habits compare to others, it creates a sense of social pressure or motivation to meet the community's recycling standards. This kind of feedback helps reinforce positive behaviors and encourages people to recycle more, as they want to align with social expectations.

For the SnackSwap project, feedback is provided directly to participants, the participants and administrators are able to see how many recyclables have been already validated. This information provides administrators with an understanding of how well the program is performing. By tracking the data, administrators can see which areas are doing well in terms of recycling and where there may be room for improvement.

Even though participants are not directly involved in the feedback process, knowing that the program is being managed effectively and that their efforts are part of the community's overall recycling goal helps motivate continued participation. The feedback system allows administrators to adjust and improve the program, making sure it remains effective in encouraging people to recycle and increasing overall participation.

1.4.4. Environmental Education

Environmental education plays a key role in shaping long-term behavior changes. According to the ERIC study, “Environmental Education and Behavioral Change: An Identity-Based Environmental Education Model” (2014), people are more likely to engage in sustainable behaviors, such as recycling, when these actions are linked to their values and identity. When individuals see themselves as someone who cares about the environment, they are more likely to act in ways that reflect this identity, like recycling regularly. This idea suggests that connecting a person’s sense of responsibility to environmental issues can lead to lasting behavior change.

The SnackSwap project makes recycling more than just a task—it becomes a part of how participants view themselves. The project encourages people to see recycling as something important, not just for the sake of waste management, but as a meaningful way to contribute to the community and help protect the environment. By offering rewards such as snacks and drinking water, SnackSwap creates a positive experience that makes recycling feel more personal

and rewarding. As participants continue to recycle and see the direct benefits, it reinforces the idea that they are playing an important role in taking care of the environment.

The monitoring also supports this by showing participants how much they have recycled over time, which helps them see their impact. When individuals can see that their actions are helping the environment and contributing to a larger goal, it strengthens their environmental identity and motivates them to keep recycling. This ongoing feedback makes recycling feel like a part of their everyday routine, connected to their sense of responsibility.

By linking recycling to personal values, SnackSwap encourages people to view it as more than just an environmental duty—it becomes a personal choice and a reflection of their commitment to the community and sustainability. Over time, this connection can lead to long-term changes in behavior, where recycling becomes a natural and consistent habit.

1.5 SCOPE AND LIMITATION

1.5.1 Scope

The project emphasizes the design and construction of a reverse vending machine capable of accepting plastic bottles and aluminum cans for recycling. The prototype is specifically designed to accommodate:

1. Transparent plastic bottles with a maximum volume of 500 mL, maximum height of 8 inches (20.3 cm), and approximate body dimensions of up to 6 cm (diameter).
2. Standard aluminum cans with similar size constraints, having a maximum height of 8 inches, a diameter of 6.6 cm, and a weight threshold ranging from 2.4 to 6 grams for aluminum cans and 2.8 to 6.1 grams for plastic bottles.

Items exceeding these dimensions or falling outside the validated weight range are rejected to maintain classification accuracy and system integrity. The prototype housing was built using

Sintra board, providing a lightweight, affordable structure for functional testing and demonstration purposes.

Reward System

The system will incorporate a component whereby users can exchange recyclable items for beverages or drinking water. The system's purpose is to promote recycling and reward the user with items supporting sustainability.

Community Awareness

The project will serve to enhance recycling awareness among the people by encouraging people to recycle plastic bottles and aluminum cans in return for rewards.

1.5.4 Limitations

1. Size and Type of Recyclable Items Accepted:

The system will only take plastic bottles and aluminum cans with precise dimensions. The height of objects to be processed should not be more than 8 inches (20.3 cm), and the capacity should not exceed 500 mL. The system will take only transparent plastic bottles and tin cans of precise size.

2. Limited Scope of Rewards:

The reward system shall be restricted to drinking water and snacks. Other reward mechanisms like cash or bigger items shall not be present in this version of the project.

3. Prototype Constraints:

Since the reverse vending machine is a prototype, its functionality shall be limited as opposed to an implemented system. It shall mostly concern functionality and user interaction, and might not have sophisticated features like large-scale distribution or automated maintenance.

4. User Testing:

The system will be tested with a minimum of 10 test users to determine usability and system effectiveness. The user interface and system performance will be improved based on feedback collected, but testing at scale or community-wide deployment is outside this project scope.

5. Material Limitation:

The machine is constructed from Sintra board, which is suitable for prototyping due to its affordability and ease of assembly. However, Sintra is not ideal for long-term durability, especially in high-traffic or outdoor environments.

Real-world deployment would require more robust materials like:

- a. Stainless steel or aluminum (for weather resistance)
- b. Polycarbonate (for impact protection)
- c. Industrial-grade acrylic for transparent sections

6. Reset Mechanism for Unused Points:

Users must redeem their reward within 2 minutes of accumulating enough points. If the user does not press a reward button within that time, the system automatically resets the accumulated points, requiring the user to start over. This is designed to avoid long wait times and keep the system available for the next user.

7. Limited Area of Deployment:

The reverse vending machine will be piloted in a designated area (e.g., a community center, school, or mall) for test and demonstration purposes. Large-scale deployment in many locations is not included in this phase.

1.6 REVIEW OF RELATED LITERATURE

The SnackSwap project, which uses reverse vending machines (RVMs) to encourage recycling by offering rewards like snacks and water in exchange for recyclable materials, builds upon existing research that shows the effectiveness of RVMs and reward-based systems in motivating recycling behaviors.

1.6.2 Reverse Vending Machines (RVMs) and Their Effectiveness

Reverse Vending Machines (RVMs) have become widely recognized tools for encouraging recycling by providing rewards in exchange for recyclables. Stojanov (2015) discusses the use of RVMs in various countries, where they have been successfully implemented to increase recycling rates. The study highlights that RVMs offer rewards such as cash, store credits, or vouchers, which motivate people to recycle. These systems are particularly effective in regions where recycling is not widely practiced.

The SnackSwap project adopts this approach by offering snacks and drinking water as rewards for recyclable materials, motivating individuals to participate in recycling. This reward-based system aims to engage individuals who may not typically recycle and raise awareness about proper waste management. Stojanov's (2015) research emphasizes the importance of rewards in recycling systems, and SnackSwap incorporates this principle to increase recycling participation and raise environmental awareness, particularly in communities with low recycling engagement.

1.6.2 Sensor Technology for Material Detection and Sorting

One of the key components of RVMs is the technology used to identify and sort recyclable materials. As noted in "Promoting Recycling in E-Commerce Platforms" (IJCRT, 2024), the use of sensors in recycling systems helps verify that items are properly sorted before

rewards are dispensed. The SnackSwap project will use a combination of ultrasonic, inductive, capacitive, IR, and weight sensors to identify recyclable items like plastic bottles and aluminum cans. Each sensor has a specific role: the ultrasonic sensor detects the presence of an object, the inductive sensor identifies metallic objects, and the capacitive sensor distinguishes between plastic and glass. The weight sensor ensures that aluminum cans meet the expected weight range.

This combination of sensors allows for precise material detection and ensures that only valid recyclables are accepted, reducing errors and improving the efficiency of the recycling process. The project relies on these non-AI-based technologies to ensure that the system remains simple, accurate, and efficient, making it accessible for communities with limited technological infrastructure.

1.6.3 Reward Systems and Social Impact

The use of rewards to encourage recycling is well-supported in the literature. Research such as "Promoting Recycling in E-Commerce Platforms" (IJCRT, 2024) highlights the effectiveness of reward-based systems in motivating people to recycle. Rewards, whether in the form of monetary compensation, goods, or services, have been shown to increase participation in recycling programs significantly. The SnackSwap project incorporates a similar reward system, offering snacks and water as rewards for recycling plastic bottles and aluminum cans. This simple rewards system not only motivates individuals to recycle but also encourages broader participation in recycling activities in communities where such programs are not widespread.

Additionally, the rewards provided by SnackSwap address the immediate needs of vulnerable populations, such as homeless individuals, children, and low-income families. By offering practical rewards like food and water, SnackSwap aims to meet the basic needs of these

groups while promoting environmentally responsible behaviors. This dual approach helps foster both social well-being and environmental sustainability.

1.6.4 Community-Based Recycling Programs

Community-based recycling programs are often more successful when they involve local engagement and provide immediate rewards. Several studies have highlighted that such programs can significantly increase participation and improve recycling rates, particularly in areas where recycling is not commonly practiced. SnackSwap aligns with this approach by integrating the monitoring that allows administrators to monitor the status of the vending machines in real-time, such as tracking water levels and trash levels. This centralized monitoring system helps ensure that rewards like snacks and water are consistently available, encouraging greater community participation. While SnackSwap does not directly track individual users, the app provides important data for administrators to ensure the proper operation of the system.

Research by Stojanov (2015) shows that community-based programs work best when there is active local involvement and when rewards or benefits are accessible. The SnackSwap system encourages recycling by offering rewards in exchange for recyclables, making it easier for the community to engage in sustainable practices. The app-based monitoring feature is essential in ensuring the system operates smoothly and remains helpful.

1.6.5 Challenges in Implementing RVM Systems

Despite the effectiveness of RVMs, there are still challenges in ensuring their proper implementation. Kumar et al. (2021) discuss the challenges of real-time processing and the need for accurate material identification. SnackSwap addresses these challenges by using a combination of sensors to verify each material before dispensing rewards. These sensors ensure

that only recyclable materials like plastic bottles and aluminum cans are accepted, which helps maintain the integrity of the recycling process.

Another challenge in implementing RVM systems is the accessibility of the machines. To maximize participation, it is important to place RVMs in convenient locations and ensure that the rewards system meets the needs of the community. The SnackSwap project aims to overcome this by ensuring that the machines are simple to use and that the rewards provided for snacks and drinking water are practical and appealing to all users, making recycling more accessible to a broader audience.

1.7 DEFINITION OF TERMS

Reverse Vending Machine (RVM) – A machine that accepts recyclable items like plastic bottles and aluminum cans and provides a reward in return. The SnackSwap project uses RVMs to encourage recycling by offering snacks and water as rewards.

Recyclable Materials – Items such as plastic bottles and aluminum cans that can be processed and reused for other purposes. These are the primary focus of the SnackSwap project to encourage recycling and waste reduction.

Monitoring – A feature in the SnackSwap system that allows administrators to track the usage of the reverse vending machines in real-time, including monitoring stock levels and the number of recyclables processed. This helps maintain the system's efficiency.

Snacks or Drinking Water – Rewards are provided to users who deposit recyclable items into the reverse vending machine. These rewards aim to motivate individuals to recycle, with the SnackSwap project targeting vulnerable populations in need of essential resources.

Recycling Awareness – Efforts to educate individuals about the importance of recycling and the positive impact it has on the environment and community cleanliness. The SnackSwap project seeks to increase this awareness by offering tangible rewards for recycling.

Plastic Bottles and Aluminum Cans – The specific recyclable materials targeted by the SnackSwap: Reverse Vending for Bottles and Cans with Monitoring for Supporting the Community project. These materials are chosen due to their high recycling potential and environmental impact.

Sensor Technology – The use of various sensors (ultrasonic, inductive, capacitive, IR, and weight sensors) in the SnackSwap project to accurately detect and verify recyclable materials before rewards are dispensed. These sensors help ensure the efficiency and accuracy of the recycling process.

Community-Based Recycling – Recycling initiatives that focus on local engagement and participation. The SnackSwap project aligns with this approach by offering rewards to encourage recycling and using technology to monitor and manage the process at the community level.

Vulnerable Populations – Groups such as homeless individuals, children, and low-income families are the target beneficiaries of the SnackSwap project. By offering rewards like snacks and drinking water, the project aims to support these groups while promoting environmentally sustainable behaviors

CHAPTER 2

METHODOLOGY

2.1 BLOCK DIAGRAM

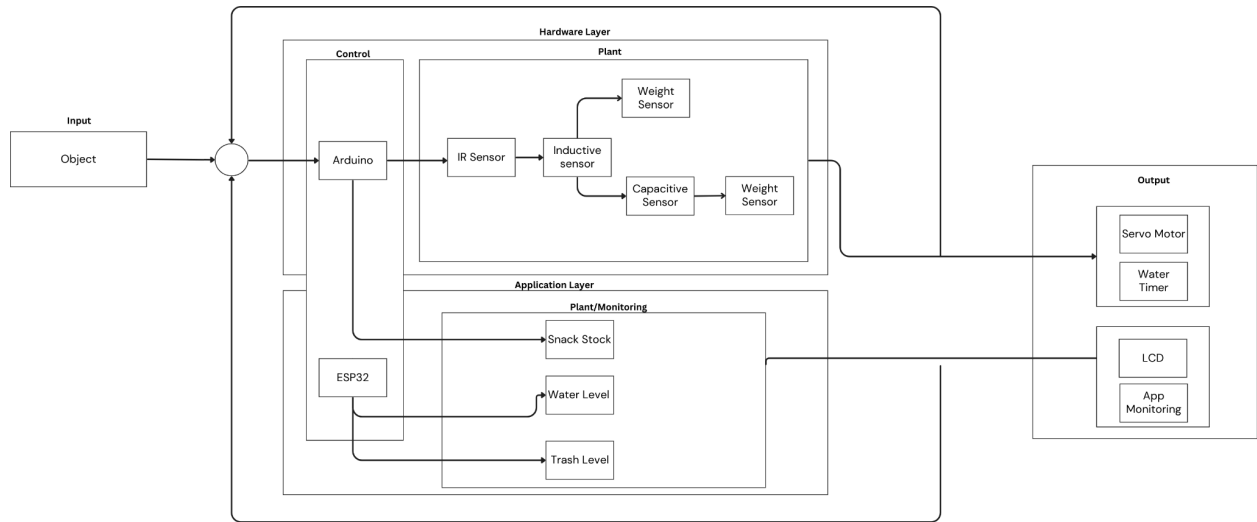


Figure 2.2.1 Block Diagram

The block diagram for the SnackSwap: Reverse Vending Machine consists of four phases: Input, Control, Plant, and Output.

The input includes an ultrasonic sensor that detects the presence of an item in the machine. Once detected, the system moves to the next phase of processing.

The Hardware Layer is divided into the Control and Plant. The Control, managed by an Arduino microcontroller, processes data from the sensors to determine if the item is valid.

The Plant contains the Inductive Sensor, which identifies metallic materials; the Capacitive Sensor, which detects non-metallic items like plastic bottles; the Weight Sensor, which verifies the weight of aluminum cans and plastic bottles. A Servo Motor in this layer manages the movement and sorting of items for validation, as well as the dispensing of snacks.

The Application Layer is responsible for monitoring and management. The ESP32 module updates real-time data on a mobile app, including water levels, and the trash bin level, to keep the system operational and efficient.

The Output includes a Servo Motor for dispensing snacks, a Water Timer for releasing water, and a Mobile App for displaying real-time data.. Based on validation results, items are either dispensed as rewards or sent to the reject bin.

2.2 PROJECT MANAGEMENT

2.2.1 Materials and Resources

1. Arduino Mega 2560 Microcontroller

The Arduino is the main processing unit, responsible for collecting data from the sensors and controlling hardware like motors and displays.

2. ESP32

ESP32 is responsible for sending data to app for monitoring.

3. Capacitive Proximity Sensor

This sensor detects non-metallic objects.

4. Inductive Proximity Sensor

The inductive sensor identifies metallic objects.

5. Ultrasonic Sensor

The ultrasonic sensor checks the trash level and is used for snack stock monitoring.

6. IR Sensor

The IR sensor initiates the start of process as well as height validation.

7. Weight Sensor

The weight sensor measures the weight of aluminum cans to confirm they are within acceptable limits.

8. Water Pump

This controls the amount of water that will be dispensed.

9. Servo Motor (Metal Gear)

The servo motor operates the flaps or doors used during the validation process as well as to release snacks from the vending machine.

10. Digital Display

The display shows whether the inserted item is accepted and provides options for rewards (snacks or water).

11. Buttons

Buttons allow users to interact with the machine and select their preferred reward.

12. Wires and Connectors

These connect various electronic components like sensors, motors, and the microcontroller.

13. Power Supply (Battery/Adapter)

Provides power to the system, ensuring continuous operation.

Material/Resource	Quantity	Cost
Arduino Microcontroller	1	800
ESP32	1	400
IR Sensor	4	200
Capacitive Proximity Sensor	1	350
Inductive Proximity Sensor	2	700
Ultrasonic Sensor	3	750
Weight Sensor	2	700
Water Pump	1	350
Servo Motor (Metal Gear)	7	1400
Digital Display	1	200
Buttons	5	50
Wires and Connectors	4 sets	1500
Power Supply (Battery/Adapter)	1	350
Auxiliary Cost	–	5000
Total Cost		12,750

Table 2.1 Materials and Resources

2.2.2 Gantt Chart

Task	Sept				Oct				Nov				Dec				Jan				Feb				March				April			
	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Project Proposal																																
Project Selection and Main Approval																																
Chapter 1 and 2																																
Video Proposal																																
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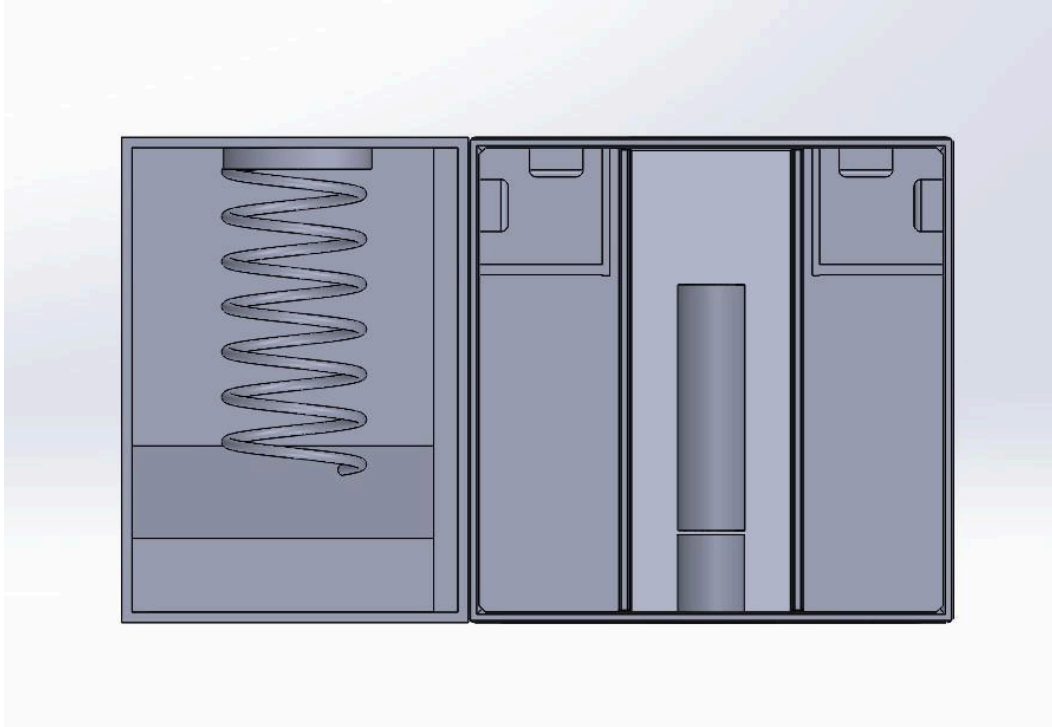


Figure 2.3.2 Top View

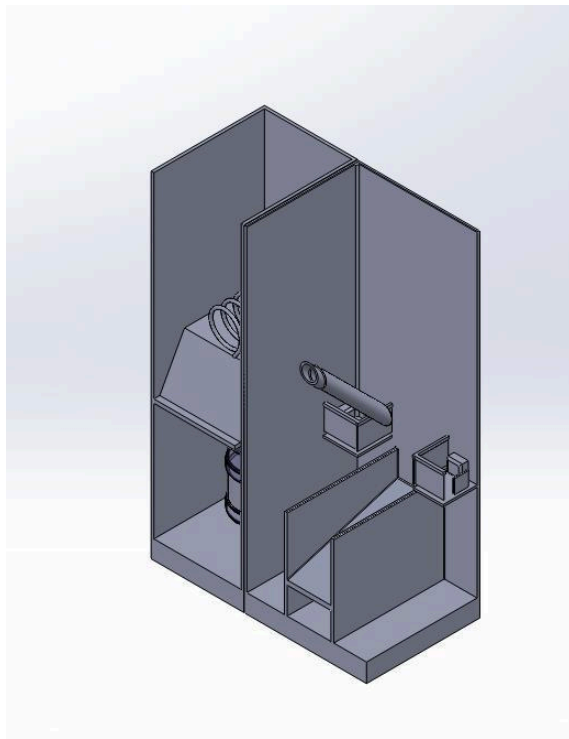


Figure 2.3.3 Right-Front View

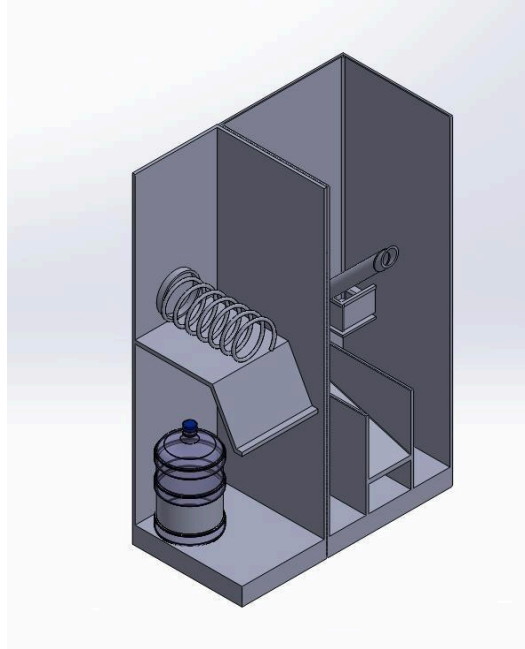


Figure 2.3.4 Left-Front View

The vending machine features an item slot where users can insert recyclable items, such as bottles and cans. The slot is positioned at an accessible height to ensure ease of use for all users. The system relies on several sensors to validate the recyclability of items. An IR sensor detects an object and triggers the start of the process, as well as it checks the height of an object, while an ultrasonic sensor is used for trash level monitoring as well as for snack stock monitoring. An inductive sensor identifies metallic items, such as aluminum cans, and a capacitive sensor for plastic bottle identification. Additionally, a weight sensor measures the weight of items to ensure they fall within acceptable ranges and prevent invalid materials from being processed.

For reward selection, the vending machine includes buttons for snacks and water. The snack button allows users to choose a snack as a reward, while the water button enables the selection of water. A digital display provides clear instructions, feedback, and reward options, improving user interaction. Accepted items lead to the reward dispensers, which include a snack

dispenser that uses a servo motor for precise item release and a water dispenser for additional reward options.

Rejected items are handled by a reject mechanism, directing them to a reject slot or trash bin to maintain the system's integrity by ensuring only valid recyclables are accepted. Accepted recyclables are collected in a trash bin, designed to store items for proper disposal or recycling. This system ensures effective recycling and reward distribution while maintaining a user-friendly experience.

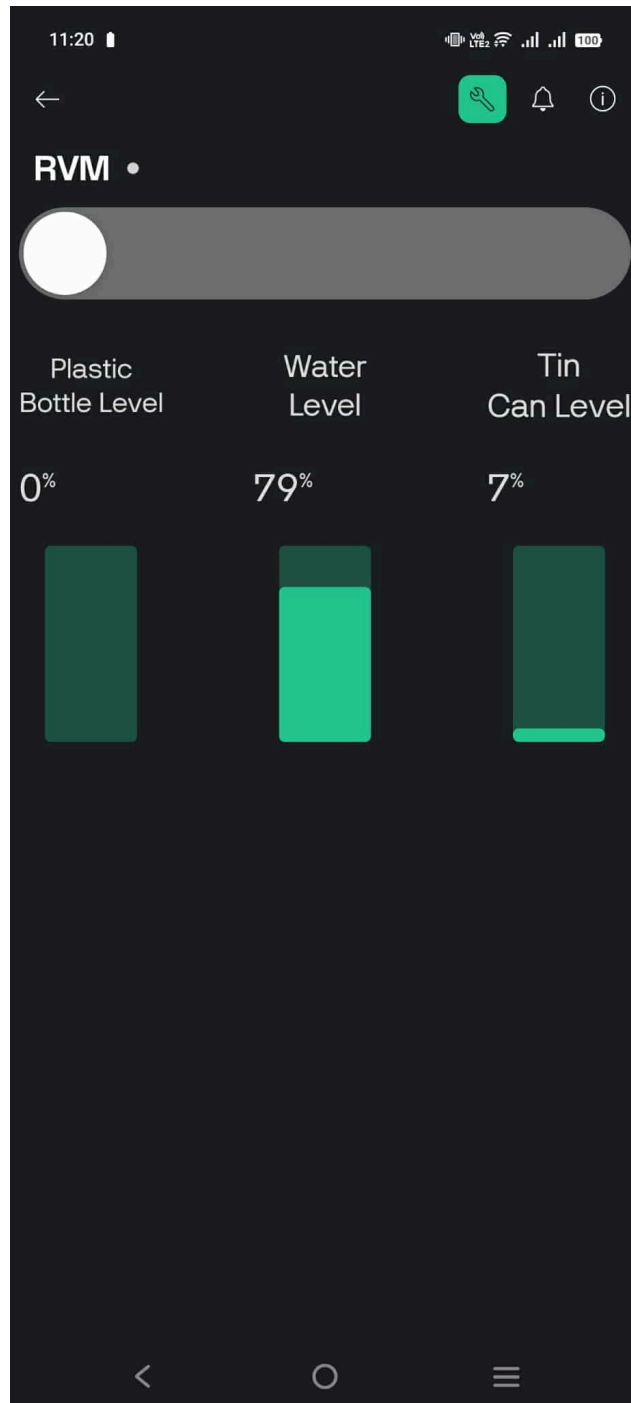


Figure 2.3.5 UI - App Monitoring

User Interface (UI) – Mobile Monitoring via Blynk App

The mobile application, developed using the Blynk IoT platform, enables real-time monitoring and control of the reverse vending machine (RVM) through an internet-connected microcontroller (such as the ESP32 or NodeMCU). The interface displays the current fill levels of various compartments: Plastic Bottle Level, Water Level, and Tin Can Level.

Each percentage value shown (e.g., 0%, 79%, 7%) represents the **current fill level** of the respective compartment. That is:

- **0%** means the compartment is **completely empty**.
- **100%** would indicate the compartment is **completely full**.

For example, in the figure:

- The Plastic Bottle Level is at 0%, meaning the bin is empty.
- The Water Level is at 79%, indicating the tank is mostly full.
- The Tin Can Level is at 7%, showing that the bin is nearly empty.

This percentage-based fill level helps users understand how close each bin is to needing maintenance or emptying, enhancing the efficiency of the system through proactive monitoring.

2.4 CIRCUIT DESIGN

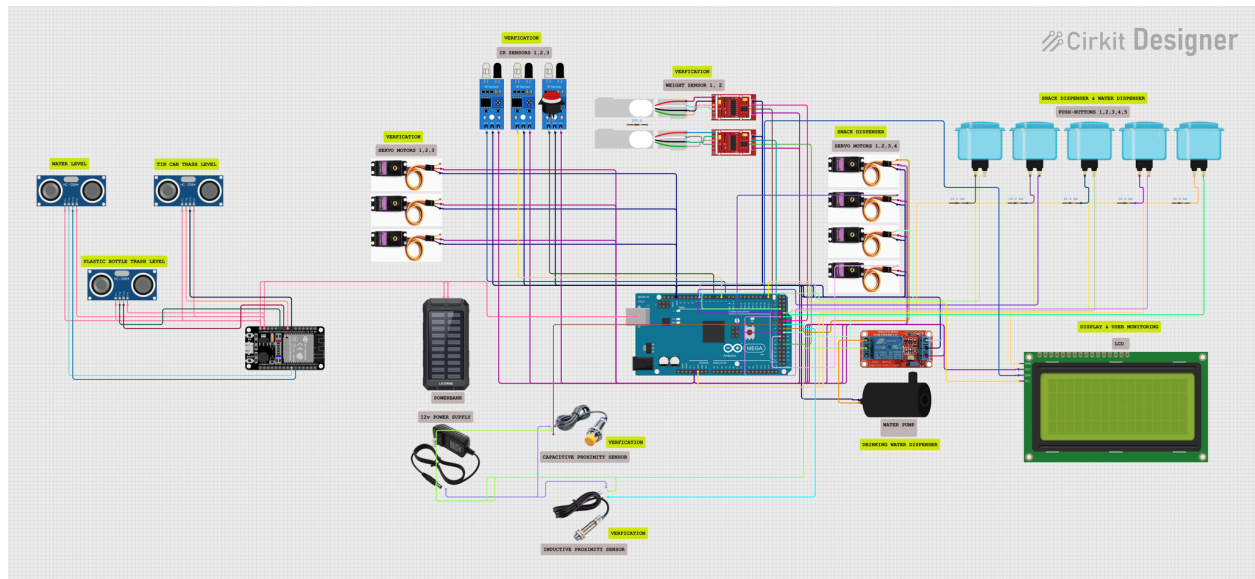


Figure 2.4.1 Circuit Board

The Arduino microcontroller is at the center of the system, handling all operations and controlling other components. The ultrasonic sensor is used to detect if the snack has already been dispensed and also for the monitoring of water level. The system then verifies the item using a combination of sensors.

The servo motor is connected to the Arduino and controls the sorting mechanism to separate plastic and aluminum cans based on material detection, as well as for dispensing snacks. IR sensors are used to start the process of validating the item.

The user buttons allow for reward selection. The snack selection buttons (labeled Snack 1, Snack 2, Snack 3, Snack 4 and Water) control the dispensing of different snacks, while the pump water button controls the dispensing of water. Once the user selects a reward, the servo motor is used to dispense the snack, while the water pump is activated to dispense water.

The total button tracks all credits input into the system, ensuring that users get their rewards once the required number of items is submitted.

2.5 PROGRAM FLOWCHART

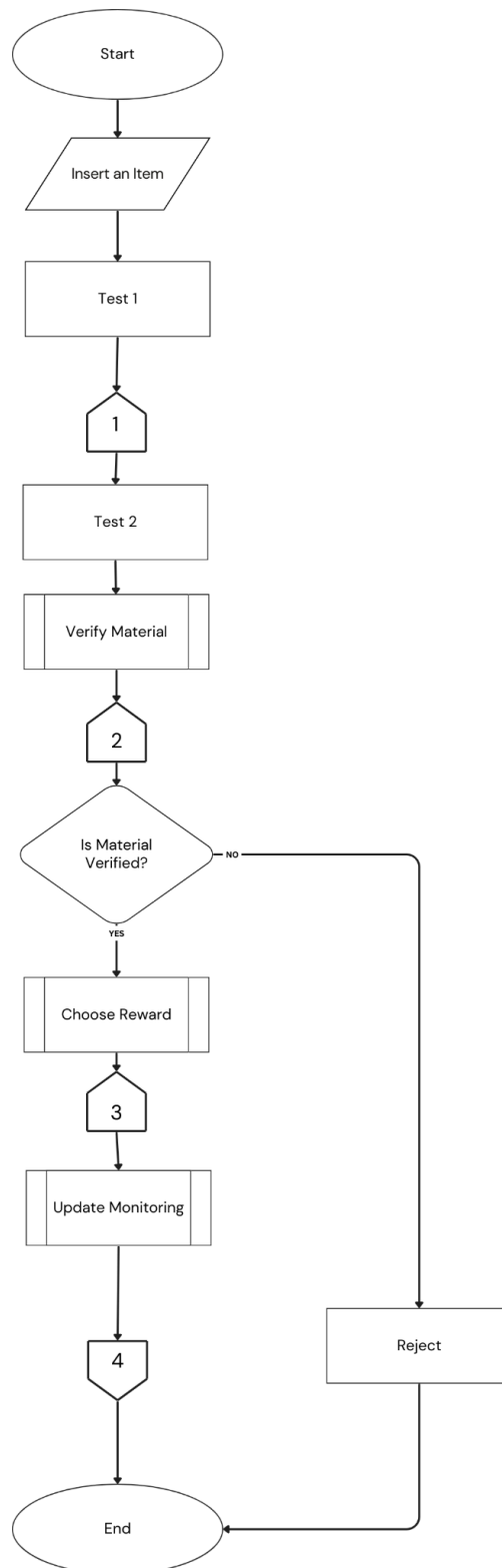


Figure 2.5.1 Program Flowchart (Object Cycle on Machine)

The process begins when an item is inserted into the machine, which is detected by an IR sensor(IR1). Once an item is detected, it then proceeds to Test 1. After which it proceeds to Test 2 wherein verification of materials happens, if the material is aluminum can or 500ml plastic bottle. If the item is not a plastic bottle or can, the flow moves to reject the item. If the material is verified, the user may choose their reward (either snack or water), this is considering if they got enough points to claim their rewards. After this, the system updates the lcd monitor, such as “Ready for Items” to proceed to next item validation. Next, the trash level is checked via ultrasonic to monitor its level, then displayed to the App monitor.

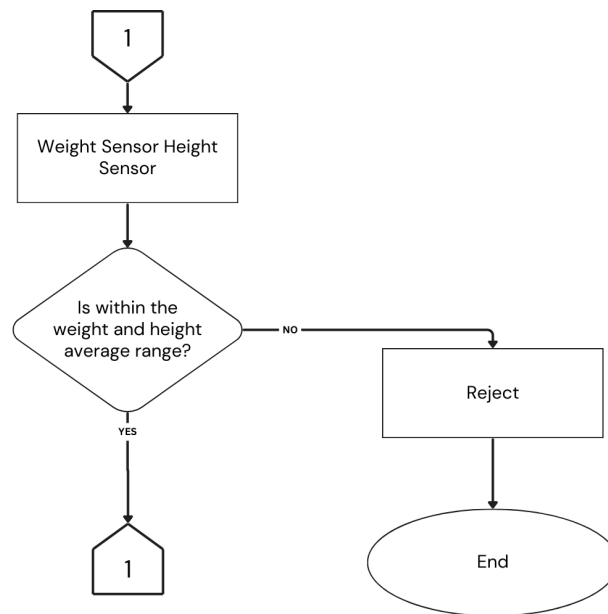


Figure 2.5.2 Program Flowchart (Test 1)

In Test 1, it evaluates the weight(Load Sensor 1) if it falls to the range of 1.5g - 8g and a maximum height(IR2) of 8 inches. With this threshold many materials are already considered as invalid because this specified requirement aims to cater aluminum cans and 500ml plastic bottles.

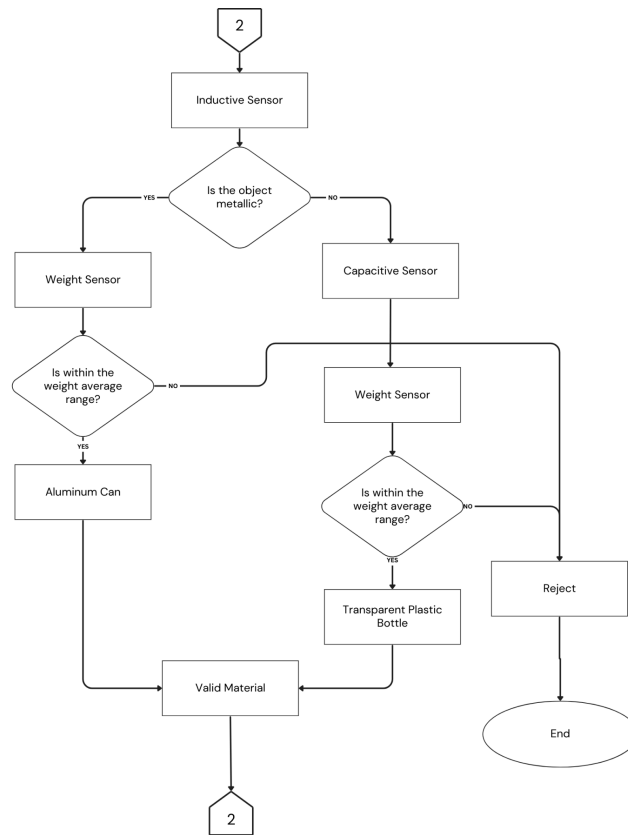


Figure 2.5.3 – Program Flowchart (Test 2 - Material Validation Process)

In Test 2, the first step checks if the object is metallic using the inductive sensor. If the item is metallic, the weight sensor then verifies if it falls within the average weight range for an empty aluminum can. If the weight is correct, the item is accepted as a valid recyclable metal and the process moves to the next step. If the item is not metallic, the system uses the capacitive sensor to check if the material is plastic or not. If the capacitive sensor detects then it means it is not a plastic bottle for a particular reason capacitive sensor only detects specific non-metallic objects and plastic bottles is not one of the objects that it detects. If it passes, the weight sensor verifies if it falls within the expected weight range for a plastic bottle. If all these checks pass, the material is confirmed as valid. If any step fails, the item is rejected.

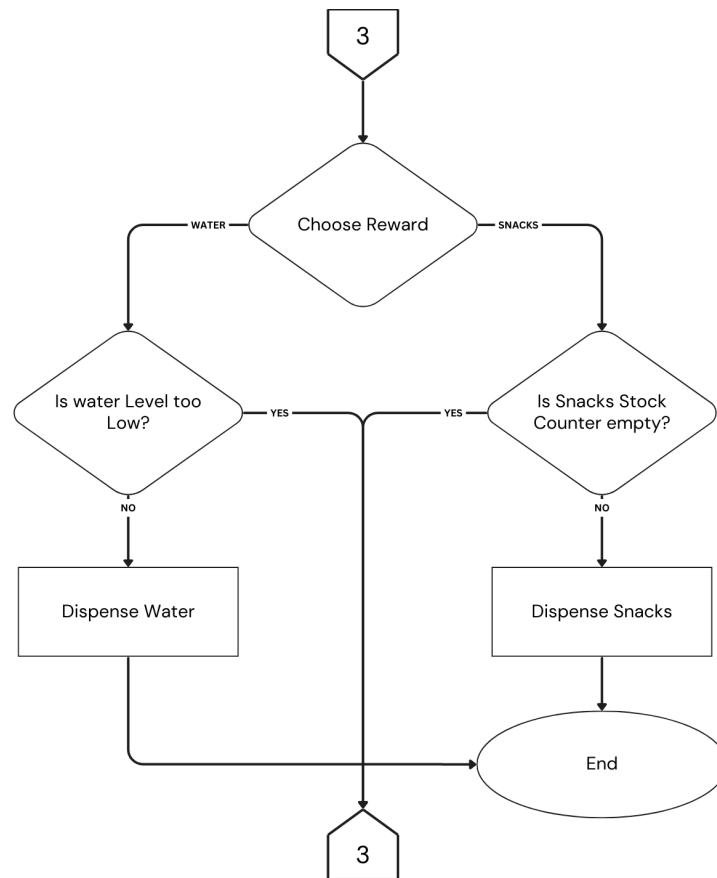


Figure 2.5.4 – Program Flowchart (Reward Selection)

After material validation, the user is given the option to select a reward. The user can only claim their rewards if they have enough points, ten(10) points for snacks and three(3) points for water. The system checks if the water level in the dispenser is low. If the water level is not low, the machine will dispense water. For snacks stock monitoring, after the user chooses their snacks the system will then dispense the snack. If the snack was successfully dispensed and falls to the snack slot the ultrasonic then stops the dispensing through a distance disruption during the fall. If the ultrasonic sensor does not detect a distance disruption after forty(40) seconds, then the lcd monitor will display that there's no snack on the said snack. Once a reward is dispensed, the process ends.

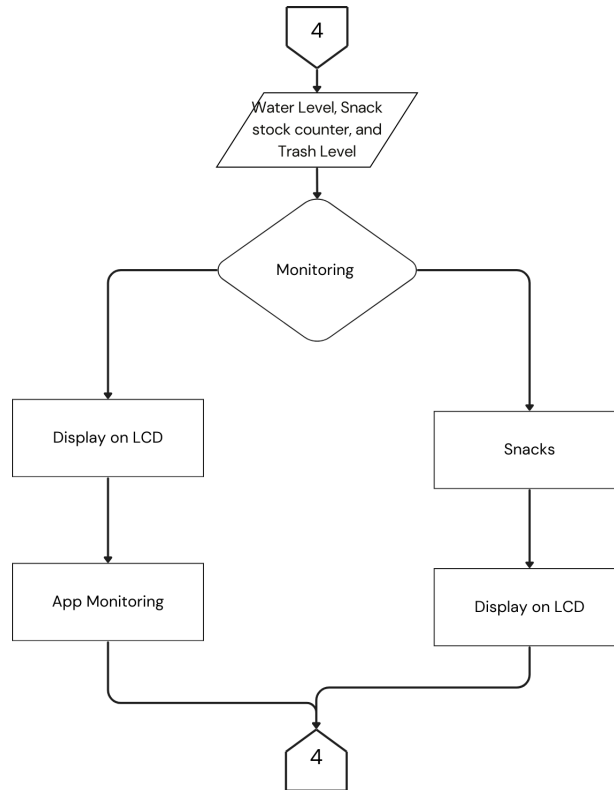


Figure 2.5.5 – Program Flowchart (Monitoring and Notifications)

The final flowchart focuses on the real-time monitoring and maintenance of the system. The ESP32 module communicates with the web application via Blynk Cloud, updating the water level and trash level. The application monitors these levels.

2.6 TESTING AND DATA GATHERING

2.6.1 Testing Strategy 1

The SnackSwap: Reverse Vending for Plastic Bottles and Cans with Monitoring for Supporting the Community will undergo testing to ensure it works as intended. The testing focuses on checking sensor performance and system response.

Data to be Gathered

1. Material Testing:

The sensors (inductive, capacitive, IR, and weight sensors) will be individually tested to evaluate their accuracy in detecting recyclable items, such as plastic bottles and aluminum cans. This test will ensure that the system correctly identifies valid recyclables and rejects non-recyclable items, maintaining the integrity of the recycling.

2. Reward Dispensing

Data will be collected to confirm that rewards (snacks or water) are dispensed correctly and on time after recyclable items are verified.

3. Trash Level Monitoring

Data will confirm if the system can accurately track trash levels and notify when the machine is full.

Importance of Data Gathering

Collecting this data is important to make sure the system works correctly and achieves its goal of encouraging recycling.

Expected Outcome

The testing should show that:

1. The sensor works correctly.
2. Rewards are dispensed smoothly.
3. The system is easy for users to operate.
4. Trash levels are tracked correctly to prevent overloading.

2.6.2 Testing Strategy2

This strategy tests the reward system logic, including its ability to correctly calculate the number of recyclable items inserted and prompt users appropriately for their rewards.

Data to be Gathered

1. Reward Calculation Accuracy

The system will be tested to ensure that it correctly counts the number of recyclable items inserted. This includes verifying that rewards are only dispensed when the required number of items is met (e.g., 10 bottles or cans).

2. Prompt Functionality

Data will track if the system accurately prompts users when the required number of items for a reward is reached. For example, displaying messages like “Points = 10. Choose your reward.”

3. Error Handling

Scenarios where fewer than the required number of items are inserted will be tested. The system should display messages such as “Not enough points” and ensure no rewards are dispensed in these cases.

4. Material Detection Accuracy:

The system will be tested to check how well the sensors (inductive, capacitive, IR, and weight sensors) detect recyclable items like plastic bottles and aluminum cans. This ensures only valid items are accepted, and non-recyclables are rejected

Importance of Data Gathering

This data is important to confirm that the reward logic works as intended. It ensures that the system only provides rewards when users meet the required conditions, maintaining the integrity of the recycling-for-reward process.

Expected Outcome

The testing is expected to demonstrate that:

1. Rewards are only dispensed when the correct number of items is inserted.
2. The system provides clear and accurate prompts to guide users.
3. No rewards are dispensed if the conditions are not met.

2.6.3 Testing Strategy 3

Monitoring Testing Strategies

This testing strategy focuses on verifying the system's ability to accurately track recyclable item deposits, monitor water levels, and manage snack stock availability.

Data to be Gathered:

1. Recyclable Item Deposit Tracking:

Test the system to ensure it accurately records the amount of plastic bottles and aluminum cans deposited by verifying if the sensors correctly track and log the number and type of items deposited, including handling bulk deposits.

2. Water Level Monitoring:

Test the system's ability to monitor the water level by checking if it updates the water level after dispensing and ensure the system triggers alerts when the water supply is low.

3. Snack Stock Availability Monitoring:

Test the system to ensure it accurately tracks snack stock by verifying that it logs snack exchanges correctly, and provides alerts when stock is low.

2.6.4 Testing Strategy 4

This testing strategy focuses on collecting user feedback to ensure the system is easy to use and meets the community's needs.

Data to be Gathered

1. User Satisfaction:

Collect feedback from users about how easy the system is to use and how satisfied they are with the rewards. Observe users interacting with the machine, such as pressing buttons, using the display, and selecting rewards.

2. Support for Test Users:

Gather feedback from at least 10 test users to see if the system helps them access rewards easily.

Importance of Data Gathering

This data helps ensure the system is easy to use, effective in encouraging recycling, and accessible to all users.

Expected Outcome

Testing should show that:

1. Users find the system easy to operate.
2. The system is accessible to all users.
3. Communication and prompts are clear and easy to follow.

CHAPTER 3

DATA PRESENTATION AND ANALYSIS

This chapter describes the tools, techniques, and procedures used to test and evaluate the SnackSwap Reverse Vending Machine (RVM). Testing was conducted to validate each subsystem: material detection, trash level monitoring, reward dispensing, and system usability.

3.1 Testing Strategy 1

3.1.1 Material Validation Using Sensors Data Presentation

Each sensor in the system capacitive, inductive, ultrasonic, and weight was tested individually and in combination to assess accuracy and reliability in detecting and classifying materials.

Test no.	Material	Material Type	Detection
1	Absolute	Plastic	x
2	C2	Plastic	x
3	Mogu-Mogu	Plastic	x
4	Summit	Plastic	x
5	Nature Spring	Plastic	x
6	Wood	Wood	✓
7	NewsPaper	Paper	✓
8	Slipper	Rubber	✓
9	Electrical Tape	Vinyl	✓
10	Rock	Rock	✓

Table 3.1. Non metallic Testing - Capacitive Sensor

3.1.2 Capacitive Sensor (Table 3.1): Detects non-metallic materials (like plastics). The capacitive sensor did not detect any of the 5 plastic bottles. It detected materials like wood, paper, rubber, etc.

1. Out of 10 test objects, 5 plastic items (e.g., C2, Mogu-Mogu, Summit) were not detected, while other materials like wood, rubber, and rock were detected.
2. This indicates that plastic bottles, being low-dielectric and non-conductive, are not detected by the sensor, offering a clear validation mechanism.
3. The items used in this test are not necessarily plastic bottles, as the goal was to check for the metal-sensing accuracy of the sensor.

Test no.	Material	Material Type	Detection
1	Coke Can	Aluminum	✓
2	Redhorse Can	Aluminum	✓
3	Sprite Can	Aluminum	✓
4	Royal Can	Aluminum	✓
5	Heiken	Aluminum	✓
6	Screwdriver	Metal	✓
7	Copper Wire	Copper	✓
8	Spray Paint	Metal	✓
9	Nails	Metal	✓
10	Aquaflask	Metal	✓

Table 3.2. Metallic Testing - Inductive Sensor

3.1.3 Inductive Sensor (Table 3.2): Identify metallic materials like aluminum cans. Detects any kind of metal samples (including cans, screwdriver, nails, copper wire).

1. All metallic samples, including aluminum cans (Coke, Sprite) and metals (screwdriver, copper wire), were successfully detected, proving that this sensor can reliably distinguish metals from nonmetals.
2. The items used in this test are not necessarily aluminum cans, as the goal was to check for the metal-sensing accuracy of the sensor.

Five test-cases were done for five plastic bottles and five other non-metal objects to get the average weight for plastic bottles. This would assist in classifying Plastic bottles against other non-metal items.

Plastic Bottle(by grams)

Test no.	Absolute	C2	Mogu-Mogu	Summit	Nature Spring
1	3.9	3.9	5.6	4.4	3.2
2	4.2	3.8	5.1	4.5	3.3
3	4.5	3.5	5.7	3.9	3.6
4	4.0	4.0	5.2	4.4	3.0
5	4.6	4.8	5.4	4.2	3.7

Other Non-Metal(by grams)

Test no.	Wood	News Paper	Slipper	Electrical Tape	Rock
1	15.1	1.7	28.2	8.6	20.6

2	14.9	1.6	27.7.	8.6	20.4
3	15.3	1.9	28.1	8.7	20.7
4	14.9	1.7	27.9	8.3	20.6
5	15.0	1.8	27.8	8.6	20.6

Table 3.3. Plastic Bottle vs Other Non-Metal Mass comparison

3.1.4 Load Sensor (Tables 3.3): Assists in classifying recyclable items based on weight. This would serve as an additional checkpoint for validation.

1. Plastic Bottles had a consistent weight range of 3.0–5.7g, while other non-metals exceeded 6.0g or below 3.0g. This allowed the system to filter recyclables using an acceptance threshold of 2.9–6.0g, consistent with item size restrictions (8” height, 14” width).

Five test-cases were done for five Aluminum bottles and five metal items (not an aluminum bottle) to get the average weight for Aluminum bottles. This would assist in classifying Aluminum bottles against other metal items.

Aluminum Can(by grams)

Test no.	Coke Can	Redhorse Can	Sprite Can	Royal Can	Heiken
1	2.7	4.5	2.7	2.2	5.9
2	2.4	5.0	3.2	2.4	4.9
3	2.3	3.9	3.3	2.3	5.3
4	2.7	4.1	2.5	2.2	5.7

Test no.	Coke Can	Redhorse Can	Sprite Can	Royal Can	Heiken
1	2.7	4.5	2.7	2.2	5.9
5	3.0	3.7	2.6	2.1	5.9

Other Metal items (by grams)

Test no.	Screw Driver	Copper Wire	Spray Paint	Nails	Aqua Flask
1	50.3	1.7	30.4	7.6	86.4
2	49.8	1.5	30.3	7.3	85.9
3	50.1	1.3	29.8	7.5	86.5
4	50.3	1.1	29.9	7.4	86.1
5	50.2	1.2	30.2	7.5	86.6

Table 3.4. Aluminum Can vs Other Metal Mass comparison

3.1.5 Load Sensor (Tables 4.4): Assists in classifying recyclable items based on weight.

1. Aluminum Cans ranged from 2.1–5.9g, while other metal items were much heavier (e.g., screwdriver ~50g). This allowed the system to filter recyclables using an acceptance threshold of 2.0–6.0g, consistent with item size restrictions (8” height, 14” width).

3.1.6 Material Detection Results

The sensors performed consistently in identifying valid recyclable items:

1. Plastic bottles were distinguished by a combination of non-detection from capacitive sensors and weight thresholding.

2. Aluminum cans were reliably identified via inductive detection and weight thresholding.
3. Items outside acceptable size or weight limits were rejected.

These findings affirm the accuracy of Test 1 and Test 2 processes, providing a high-confidence validation of item classification.

3.2 Testing Strategy 2

3.2.1 Trash Level Detection

Test no.	Actual Trash Level (%)	Level Detected (%)	Detected Accuracy	App Monitoring Displayed	Response Time (s)
1	15	18	97%	✓	8.6
2	25	29	96%	✓	10.5
3	50	52	98%	✓	12.4
4	65	70	95%	✓	9.7
5	40	43	97%	✓	16.4
6	80	82	98%	✓	11.5
7	90	92	98%	✓	10.9
8	30	33	97%	✓	14.2
9	45	48	97%	✓	13.6
10	20	21	99%	✓	9.9

Table 3.5. Trash level Detection Accuracy

Ultrasonic Sensor (Table 3.5) was used to detect when the trash bin was nearing full.

1. The sensor showed a 95–99% detection accuracy against a 90% actual fill level, triggering in the app monitoring within an average of 11.77 seconds

Trash Monitoring Accuracy

The ultrasonic sensor accurately measured trash levels near full capacity:

1. With near-perfect accuracy ($\geq 95\%$), the app monitoring reliably monitors the trash level.
2. This aligns with the system's goal of maintaining continuous operability and minimizing downtime.

3.2.2 Reward Dispensing and Point Accuracy

This test evaluates how accurately the system:

1. Detects recyclable items (plastic bottles and cans.
2. Assigns the correct number of points
3. Dispenses the appropriate reward based on user input

Each valid item (bottle or can) = 1 point

Redemption thresholds:

1. 10 points = Snack
2. 3 points = Water

Test No.	No. of Items Inserted	Items Inserted	No. of valid Items Detected	Expected Points	System Points	System Points Accuracy	Button Pressed for Reward	Actual Reward Dispensed	Reward Dispensed
1	10	2- C2 2 - Nature Spring 1 - Mogu-mogu 2 - Absolute 1 - Summit 1 - Red Horse Can 1 - Royal Can	10	10	10	100%	Button 2	SkyFlakes	✓

2	3	1 - Coca Cola Can 1- C2 1 - Absolute	3	3	3	100%	Button 5	Water	✓
3	10	1 - Heineken Can 1 - Red Horse Can 1 - Coca Cola Can 1 - Sprite Can 1 - Royal Can 1 - Summit 1 - C2 2 - Nature Spring 1 - Mogu-mogu	10	10	10	100%	Button 3	Presto	✓
4	10	1 - Nature Spring 1 - Sprite Can 1 - Royal Can 1 - Heineken Can 1 - Coca Cola Can 1 - Coca Cola Plastic 1 - Summit 1 - Absolute 2 - C2	10	10	10	100%	Button 4	Presto	✓
5	10	2 - Absolute 2 - Nature Spring 2 - C2	10	10	10	100%	Button 1	SkyFlakes	✓

		1 - Heineken Can 1 - Red Horse Can 1 - Royal Can 1 - Coca Cola Can							
6	6	1 - Coca Cola Can 1 - Royal Can 1 - Sprite Can 1 - Absolute 1 - C2 1 - Nature Spring	6	6	6	100%	Button 5	Water	✓
7	13	1 - Red Horse Can 1 - Royal Can 1 - Coca Cola Can 1 - Sprite Can 1 - Heineken can 2 - C2 2 - Nature Spring 1 - Coca Cola Plastic 2 - Absolute 1 - Summit	13	13	13	100%	Button 1 Button 5	SkyFlakes Water	✓
8	13	2 - Absolute 1 - Mogu-mogu 1 - Summit 1 - C2	13	13	13	100%	Button 3 Button 5	Presto Water	✓

		2 - Nature Spring 1 - Coca Cola Plastic 1 - Coca Cola Can 1 - Sprite Can 1 - Heineken can 1 - Royal Can 1 - Red Horse Can							
9	13	2 - Nature Spring 3 - C2 1 - Summit 2 - Absolute 1 - Royal Can 1 - Sprite Can 1 - Coca Cola Can 1 - Heineken Can 1 - Red Horse Can	13	13	13	100%	Button 4 Button 5	Presto Water	✓
10	13	1 - Coca Cola Plastic 1 - C2 1 - Nature Spring 2 - Absolute 1 - Summit 1 - Heineken Can 1 - Red Horse Can	13	13	13	100%	Button 2 Button 5	SkyFlakes Water	✓

		1 - Royal Can 1 - Coca Cola Can							
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Table 3.6 Reward Computation Accuracy

1. Each valid recyclable earned points: 1 points for a plastic bottle and 1 points for a can.
2. Reward Options:
 - a. Snacks and Water
3. In 10 trial runs (Table 3.6), the system showed:
 - a. 100% detection and classification accuracy
 - b. All rewards were dispensed correctly based on button selection and point eligibility

Reward System Performance

1. All inserted valid items yielded correct point allocations.
2. Rewards were dispensed with 100% reliability, proving the reward system logic is accurate.

3.2.4 Error Handling

Test No.	Items Inserted	Response Time(s)	Error Message Displayed
1	330ml Pale Pilsen	14.56	✓
2	Big Wood	10.3	✓
3	Coke 8'oz	11.5	✓
4	Spray Paint	15.6	✓
5	WD-40 with content	14.8	✓

Table 3.7 Error Handling Accuracy

1. Items such as big wood, spray paint, and Pale Pilsen were rejected with proper error messages.
2. This test ensured system robustness against invalid inputs.

Error Handling Reliability

All invalid items tested were successfully rejected with proper error messaging. This confirms the system's ability to:

1. Protect against misuse
2. Maintain reward integrity
3. Enhance user trust

3.3 Testing Strategy 3

3.3.1 Unit Testing

Test Users	No. of Items Inserted	System Points	Dispensed Reward	End-to-End System response Time (m:s)
1	10	10	Snacks	4:48:56
2	3	3	Water	1:32:08
3	10	10	Snacks	4:42:87
4	10	10	Snacks	4:45:47
5	10	13	Snacks and Water	4:54:76
6	6	6	Water	2:18:35

7	13	13	Snacks and Water	6:30:56
8	13	13	Snacks and Water	6:45:21
9	13	13	Snacks and Water	6:39:71
10	13	13	Snacks and Water	6:50:35

Table 3.8. User Unit Testing

System usage by 10 test participants revealed:

1. Clear tracking of existing points and earned rewards
2. Correct rejection when points were insufficient
3. Smooth interaction with reward buttons and display
4. Response times between 1:32:08 and 6:50:35 minutes depending on user interaction

User Experience

Participants were able to:

1. Interact with the machine confidently
2. Understand the point system and reward options
3. Complete transactions without confusion

3.4 Testing Strategy 4

A survey was conducted with 10 users who participated in the User Unit Test to gather feedback on the system's usability and its effectiveness in meeting community needs.

Metric	Average Score (out of 5)
Ease of Use	4.3
Reward Satisfaction	4.5
Awareness of Program	4.4
Encouraged Recycling	4.7
Environmentally Beneficial	5.0

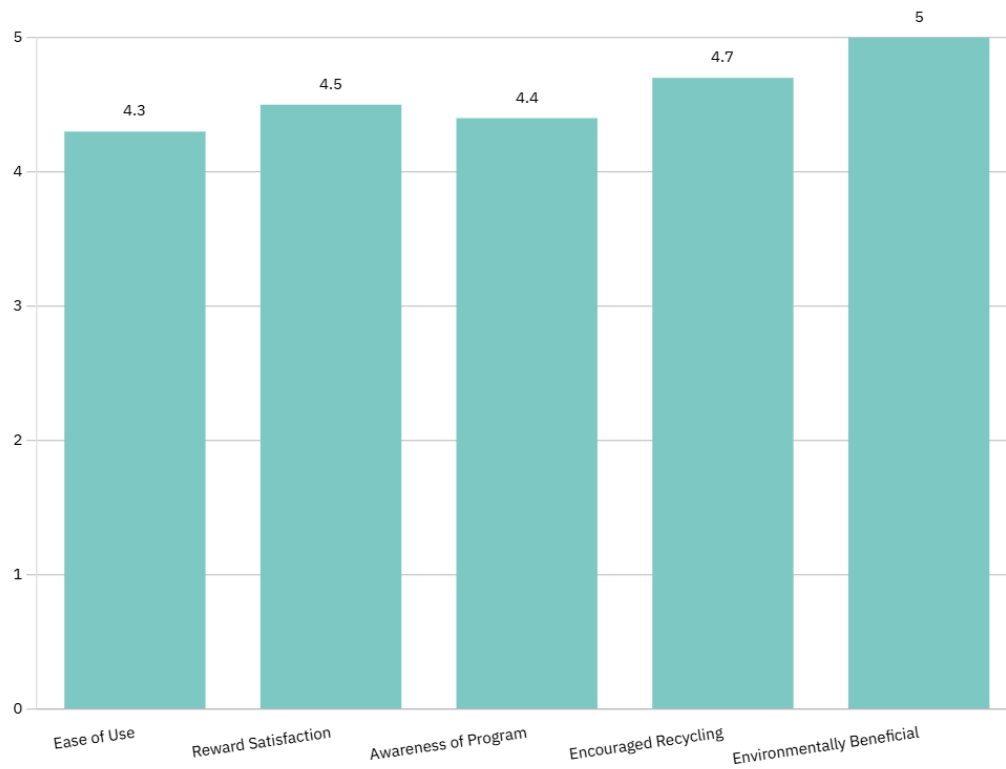


Figure 3.10 User Satisfaction Bar Graph

1. Ease of Use (4.3):

Majority of users found the system easy to operate, indicating a user-friendly interface.

Minor adjustments could enhance elderly accessibility based on the lower score (3) from the 50–65 age group.

2. Reward Satisfaction (4.5):

Users were generally pleased with the snack/water rewards, suggesting the current reward options are well-received and motivating.

3. Awareness of Program (4.4):

High scores reflect good communication of the project's goals. However, some users (e.g., Test User 2 and 6) were less informed, indicating room for improved awareness strategies.

4. Encouraged Recycling (4.7):

The system strongly reinforces recycling behavior—a clear success aligned with your theoretical framework (operant conditioning and environmental education).

5. Environmentally Beneficial (5.0):

All users rated the machine as highly beneficial to the environment, showing strong community approval of your project's sustainability goals.

CHAPTER 4

CONCLUSION AND RECOMMENDATION

4.1 Conclusion

The SnackSwap: Reverse Vending Machine for Plastic Bottles and Cans with Monitoring for Supporting the Community successfully met its design objectives of encouraging recycling, ensuring accurate validation of recyclable materials, and providing a functional reward-based incentive system.

The system demonstrated the following outcomes:

1. Accurate classification of recyclable items using a combination of capacitive, inductive, and weight sensors.
2. Reliable reward dispensing, with 100% accuracy in item detection, point allocation, and reward output.
3. Effective monitoring of trash bin levels and inventory using ultrasonic sensors, ensuring system readiness and timely maintenance.
4. User-friendly experience, based on positive feedback and successful navigation of all reward and validation processes by test participants.

These results support the theoretical framework laid out in Chapter 1, particularly in terms of behavioral reinforcement, community engagement, and environmental education. The SnackSwap machine effectively translated theory into practice, making recycling an engaging and rewarding experience.

4.2 Recommendations

To further improve and expand the system, the following recommendations are proposed:

1. Expand Material Detection

- a. Include validation for colored plastics and glass bottles, expanding the scope of recyclables.
 - b. Integrate additional sensors or AI-based image recognition for broader material classification.
- 2. Refine Weight Calibration
 - a. Add adaptive weight calibration to adjust thresholds automatically based on environmental conditions or object patterns.
- 3. Improve User Feedback System
 - a. Implement a visual display of accumulated points and past contributions for the user.
 - b. Offer multilingual support for broader accessibility.
- 4. Enhance Reward Options
 - a. Expand the reward system to include e-vouchers, coupons, or store credits in future community rollouts.
- 5. Wider Deployment and Data Logging
 - a. Pilot the machine in public areas such as schools, malls, or barangay centers.
 - b. Include cloud integration for remote monitoring and data analysis of recycling trends.
- 6. Sustainability Education Integration
 - a. Include infographics or videos on the display to educate users about environmental impact, increasing awareness alongside behavioral reinforcement.

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